

Practical Implementation of an OFDM Based Active Radar System for Target Classification



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Declaration

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Acknowledgements

At the start of my engineering degree, a friend of my parents, who happens to be an engineer himself, heard that I had been accepted into the electrical engineering faculty at UCT. I asked him for any advice that he might have for me, anything that could potentially help me through the four years that lay ahead of me. He said one thing to me, ‘Make a group of close engineering friends, stick together, and you’ll get each other through’. This was some of the best advice that I have ever received. To everyone in the engineering friend group, getting to know everyone over the last few years has been one of the coolest experiences in my life. Thank you for all the memories made. I cannot wait to see what interesting things everyone gets up to in the coming years.

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Abstract

Every day without fail, all around the world, there are thousands of weather balloons that are launched into the atmosphere for meteorological observation and research applications. These weather balloons are all comprised of a latex balloon that is filled with helium, and a radiosonde unit that hangs below the balloon itself. This radiosonde module plays a very important role in the measurement of atmospheric conditions within which the balloon resides, as it contains the numerous electrical components that give it the ability to measure characteristics of the atmosphere and the position in which the radiosonde finds itself in, as well as transmit those readings back to a ground station.

Currently, there are only a handful of companies around the world that design, manufacture, and sell radiosonde modules. These companies essentially have a monopoly on the entire market. One of the hindrances to companies/individuals designing and manufacturing their own radiosonde modules is the complicated process of transmitting the recorded information back to a ground station. Since the companies that manufacture the radiosonde modules have closed designs on all of their components, information is not widely available as to the exact inner workings of their designs for their transmission systems, other than the broad modulation schemes. It is therefore of paramount importance that alternative transmission schemes to the schemes that are employed by the few companies that manufacture radiosondes are explored if alternative designs to the mass-manufactured models are desired.

This report outlines research into an alternative transmission system based on the [Long Range \(LoRa\)](#) communication system. While LoRa is still a proprietary communication system, it is widely available due to its popularity among hobbyist communities, as well as its expanding footprint within the [Internet of Things \(IOT\)](#) community of devices. Reasons for popularity within these communities include the long-range transmission capabilities of the communication protocol, as well as its low cost, which makes it an excellent choice for the disposable radiosonde modules that are used for atmospheric observations.

A transmission and receiver system was designed, simulated, and built, based around the LoRa transmission protocol. Tests were then performed on the system to determine the optimal LoRa parameters for the usage of the protocol in radiosonde modules. From testing and simulation data, the best bandwidth was determined to be 62.5kHz, and the best spreading factor was determined to be 11.

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Acronyms

AT&T American Telephone and Telegraph Company

FDM Frequency Division Multiplexing

IOT Internet of Things

LoRa Long Range

OFDM Orthogonal Frequency Division Multiplexing

RADAR Radio Detection and Ranging

SDR Software Defined Radio

TDM Time Division Multiplexing

TDMA Time Division Multiple Access

Chapter 1

Introduction

This report outlines the design and implementation of a single pulse classification [Radio Detection and Ranging \(RADAR\)](#) that uses an [Orthogonal Frequency Division Multiplexing \(OFDM\)](#) based signal with a flat top spectra. A literature review of RADAR and OFDM is presented, followed by the literature concerned with using OFDM signals in classification RADARs. Next, the design of the surrounding system to transmit and receive data, process the data, and analyse the results is presented. Following this, an experimental design is laid out and explained in detail. Next a simulation is performed and then the real world testing is performed. The results and comparison of the simulation and real world experiment are then discussed.

1.1 Background

RADAR technology has unlocked the ability for humans to be able to see the world around us with a new perspective. This ability to be able to see objects and more specifically, to see information such as position and velocity about these objects has helped with the development of devices and services that are of paramount importance to society. All the current use cases of RADAR do however lack the ability to determine exactly what object is being seen by the overall RADAR system. That is where the use of the OFDM waveform in a RADAR system becomes useful.

1.2 Problem Statement

Current RADAR technology works very well when determining the distance and direction of an object, however, it does not cope well with specific object classification tasks. One of the reasons for this is that current RADAR utilize either a pulse, continuous wave, or chirp waveform, which does not allow for object detection through statistical analysis due to the simple nature of the waveform. This is a drawback of using RADAR systems, since if object classification is desired, then a combination of radar and some other technology such as cameras would have to be used which increases the cost and complexity of the system as a whole.

1.3 Objectives

The goal of this project is to build a RADAR system that can simultaneously act as a normal RADAR which can determine the distance and direction of an object, as well as classify the specific object that it is looking at when compared to other example objects. The aim is to implement an OFDM based waveform into a RADAR system that would give the flexibility to analyze the returned signal statistically and determine what exactly the object is. The goal is to make a non-realtime system, that can distinguish between three different objects.

1.4 System Requirements

Since this system is very experimental in nature, the requirements are quite broad as of writing this, at the beginning of the project **OBVIOUSLY NEED TO CHANGE THIS EVENTUALLY**. The system needs to perform the following tasks.

1. Be able to operate as a normal RADAR. This entails determining range, azimuth, and elevation.
2. Be able to classify between three separate reflectors/scatterers.

1.5 Scope and Limitations

The scope of this project has been set to not include design of RADAR hardware specifically. This is because design of RADAR hardware is a complex task that would probably encompass an entire masters project in and of itself. Therefore, the RADAR hardware portion of the project will be handled by a [Software Defined Radio \(SDR\)](#). This extrapolates the complex RADAR hardware design aspect away from the project and allows me to focus on the data collection and processing aspect of the project.

Chapter 2

Literature Review

This chapter aims to add context to the entire project by exploring the existing literature on RADAR systems, the OFDM modulation scheme, and the combination of the two within a single system in order to achieve the objective of object classification. In order to get to a level where the topic of object classification can be discussed, one must first delve into the fundamental operation of RADAR and OFDM based systems. Once one has a strong understanding of these basics, the combination of the two can be discussed in regard to object classification.

2.1 RADAR

2.1.1 Brief RADAR History

The basic idea of RADAR was experimented with in the classical electromagnetic radiation experiments of the 1880s performed by Heinrich Hertz [1]. These experiments set out to prove the previous theoretical work done by James Clerk Maxwell. Maxwell came up with a formula that described electromagnetic fields, and he hypothesized that this formula would apply to both light and radio waves since they were both examples of electromagnetic radiation, just at very different frequencies. The work done by Maxwell led to the discovery that electromagnetic waves can be reflected by metallic objects and refracted by a dielectric medium, just like light can [1]. This work did not go unnoticed, and a patent for an ‘an obstacle detector and ship navigation device’ was issued in 1904 to Christian Hülsmeyer, a German engineer [1]. Work on RADAR then slowed down due to a lack of interest from governmental authorities until the early 1930s, when the development of long-range bombers prompted the major countries of the world to develop technologies which could detect these hostile aircraft.

2.1.2 RADAR Basics

The fundamental operation of a simple single pulse RADAR system relies on three physical fundamental properties of the universe [2] which are listed below:

1. **Reflection of electromagnetic waves:** This rule describes how a radio wave will be reflected back towards a transmitter of said radio wave if the object is placed directly in front of the transmitter. This reflection indicates that the object is in front of the transmitter.
2. **Electromagnetic energy travels through air at a constant speed:** This rule describes how the electromagnetic waves transmitted by an antenna propagate through the transmission medium, usually air for our earth’s use cases, at roughly the speed of light, c . This velocity c equates to $299,792,458 \text{ m.s}^{-1}$
3. **Electromagnetic energy normally travels through space in a straight line:** This describes how the energy transmitted by an antenna in a radar system propagates through space in a

straight line. This is important because it means that the energy of the RADAR system can be directed in a specific direction by the antenna system, which means that the reflecting object's direction (azimuth and elevation) can be determined.

Using the above-mentioned physical properties, the distance, azimuth, and elevation of the object that is being looked at by the radar system can be determined.

Since RADAR utilizes the electromagnetic spectrum, there are distinct advantages concerning the use of RADAR systems [2]. The main advantage is that, unlike humans, RADAR systems can operate day and night, 24/7, over far distances, in good or bad weather. Compared to human eyes, this is much better. The above example of a RADAR system is that of a simple single pulse RADAR. In the next section I want to outline the other type of RADAR system that I might use in my OFDM system, a pulsed RADAR system.

2.2 OFDM

2.2.1 Brief OFDM History

The idea of OFDM transmission techniques has been around for many years, however, it was never implemented in communication channels until the appearance of mass market applications coincided with the availability of efficient software and electronic technologies [3]. Before OFDM however, there was [Frequency Division Multiplexing \(FDM\)](#).

2.2.1.1 FDM

FDM system innovation was built out of necessity by telegraph companies hoping to increase the amount of users that could utilize their transmission lines so as to increase profits from their systems [3]. Transmission companies needed a method to have multiple messages being sent on the same transmission line at one time. Other methods had been used such as [Time Division Multiplexing \(TDM\)](#), and [Time Division Multiple Access \(TDMA\)](#). These methods were however limited by burst speed which was itself limited by inter-symbol interference. Around the 1920s, FDM became the main carrier system for telephone companies. The first major company to deploy the system was [American Telephone and Telegraph Company \(AT&T\)](#). This first system used 8kHz subchannel spacing with 4kHz channel size. This form of transmission system later developed into systems that had more sub carrier channels with less subcarrier spacing. There were however some drawbacks to FDM, which are highlighted below:

1. **Waste of frequency:** Due to the large subcarrier spacing, FDM wasted a lot of precious frequency spectrum to these so-called 'guard' bands. As the frequency spectrum became more crowded throughout the years, these guards band became a liability because spectrum access became so crowded.
2. **Using different modulators for the various sub-channels:** In the old systems, each sub-channel had a separate modulator in the up-mixing process. This multiplication process became tedious once there were lots of sub-carriers.

To compensate for the waste of frequency issue, the concept of OFDM was introduced, where each subcarrier overlapped except they had non-interfering spectra [3].

2.2.1.2 OFDM

As mentioned before, **OFDM** is a method of data transmission that breaks information into bit chunks that are sent in parallel over multiple channels [4]. The OFDM transmission scheme builds upon FDM where multiple channels are created over a specific bandwidth, and bits of information are sent on each channel in parallel. The main difference between FDM and OFDM is therefore the orthogonality of the OFDM signal. Pictured below in figure 2.1 is an OFDM, and FDM based waveform. As can be seen, all the peaks of the OFDM waveform have zero interference with any other part of the signal, whereas, with the FDM waveform, the peaks have significant interference from the other channels since the channels are not spaced orthogonally in regard to each other.

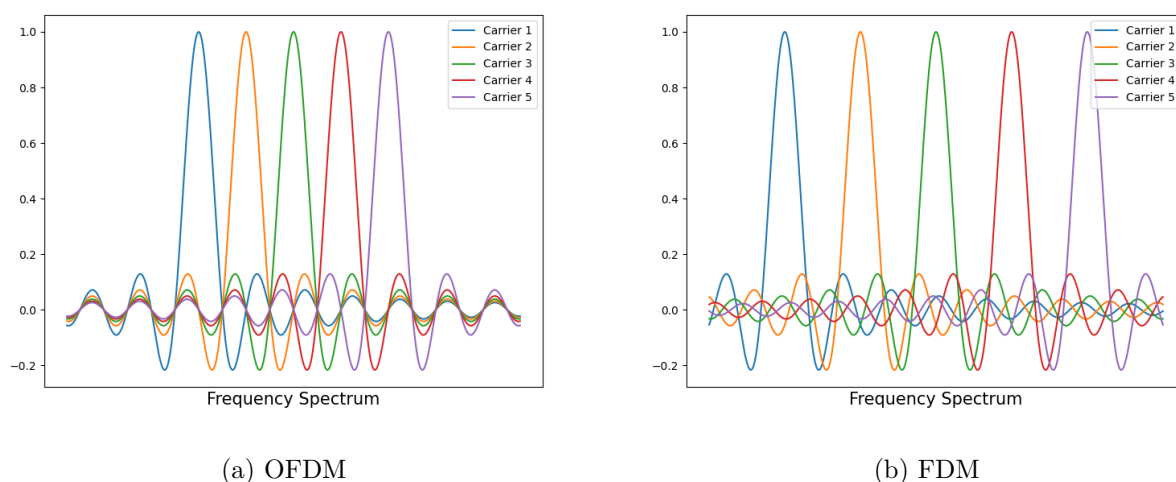


Figure 2.1: OFDM and FDM Waveforms

The above figure gives a broad idea of what goes into an OFDM-based signal, however in practice, the signal used will have vastly more complexity added on concerning error correction and data formats in general. The base signal however will still revolve around the idea of orthogonal carriers.

2.3 Fresnel Zones

2.4 LoRa Theory of Operation

2.4.1 Frequency Shift Keying

2.4.2 The Chirp

2.4.3 Bandwidth

2.4.4 Spreading Factor

2.4.5 LoRa Data Packets

2.4.6 LoRa Transmission Parameters

2.4.6.1 Symbol Rate and Symbol Time

2.4.6.2 Preamble Time

2.4.6.3 Payload Time

2.4.6.4 Packet Time

2.5 Critique of Literature

Chapter 3

System Design

[2]

3.1 Requirement and Specification Analysis

3.1.1 System Requirements

3.1.2 System Specifications

3.2 Modules and Sub-modules

3.3 Transmission Module Component Selection

3.4 Receive Module Component Selection

3.5 Hardware Interface Design

3.6 Software Design

3.7 PCB Design

Chapter 4

Validation and Simulation

[2]

4.1 Validation

4.1.1 Transmitter Validation

4.1.1.1 Power Module

4.1.1.2 Microcontroller

4.1.1.3 Sensor Modules

4.1.1.4 LoRa module and Display and Control Module

4.1.2 Receiver Validation

4.2 Simulation

Chapter 5

Methodology

[2]

5.1 Research Design and Approach

5.2 Data Collection

5.3 Shorter Range Tests

5.4 Long-Range Test

Chapter 6

Results

This section outlines the results of the experiments performed that are outlined in section 5. Discussion and analysis of results will not form part of this chapter, but rather the raw data captured will be presented for all the important tests that took place. [2]

6.1 Same Room Test

6.2 100m Test

6.3 300m Test

6.4 500m Test

6.5 20km Test

6.6 Average RSSI Values with Distance

Chapter 7

Discussion

[2]

7.1 RSSI Values with Distance

7.2 Overall Testing Performance

7.3 Determining the Optimal Packet Structure

7.4 Determining the Optimal LoRa Parameters

Chapter 8

Conclusion and Recommendations

[2]

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Chapter 9

Appendix

[\[2\]](#)

9.1 Full Transmitter Circuit Schematic

9.2 Full Receiver Circuit Schematic

9.3 Long Range Test Interference

9.4 GitHub Link

9.5 Ethics Clearance